BookLeaf Publishing

Author Support & Communication Portal

Technical Assignment Submission & Setup Guide

Candidate: Vishu Vatsay
Role: Full Stack Developer
Date: October 2026
Stack: React 18 • Node.js • MongoDB

1. Project Submission & Live Endpoints

Resource	Link / Access URL	Description
GitHub Repository	`https://github.com/vishu9520/BookLeafAssignment`	Full source code (Client, Server, Seed, Docs)
Live Backend API	`https://bookleafassignment-0oxm.onrender.com`	Production REST API with CORS enabled
Interactive Swagger Docs	`https://bookleafassignment-0oxm.onrender.com/api-docs`	Full OpenAPI specification UI with live 'Try it out'
API Health Check	`https://bookleafassignment-0oxm.onrender.com/api/health`	Service health & active AI engine status probe

2. Prerequisites

- **Node.js:** v18.0.0 or higher (v20+ recommended)
- **npm:** v9.0.0 or higher
- **MongoDB:** MongoDB Atlas connection string (or local MongoDB on port 27017)
- **Git** installed

3. Step-by-Step Local Setup

Step 1: Clone the repository and navigate into the root directory

```
git clone https://github.com/vishu9520/BookLeafAssignment.git
cd BookLeafAssignment
```

Step 2: Install dependencies for both Server and Client

```
npm run install:all
```

Step 3: Configure Environment Variables

```
cp server/.env.example server/.env
```

```
PORT=5000
MONGODB_URI=mongodb+srv://:@.mongodb.net/bookleaf?retryWrites=true&w;=majority
JWT_ACCESS_SECRET=your_jwt_access_secret_super_secure_2026
JWT_REFRESH_SECRET=your_jwt_refresh_secret_super_secure_2026
JWT_ACCESS_EXPIRES_IN=15m
JWT_REFRESH_EXPIRES_IN=7d
# AI LLM Engine (Optional: built-in heuristic fallback automatically activates if omitted)
GROQ_API_KEY=your_groq_api_key_here
GROQ_MODEL=qwen/qwen3.8-27b
LLM_PROVIDER=groq
CLIENT_URL=http://localhost:5173
NODE_ENV=development
```

4. Database Seeding (Pre-populate Official Dataset)

Run the automated seed command to import all 10 authors, 18 books, and initial tickets into MongoDB:

```
npm run seed
```

5. Running the Application (Local Development)

Launch backend and frontend in two separate terminal windows from the root directory:

Terminal 1: Backend Express Server & WebSockets `npm run server` • Listening on: **http://localhost:5000** • Swagger Docs: **http://localhost:5000/api-docs** • Health Check: **http://localhost:5000/api/health**	**Terminal 2: Frontend React 18 SPA (Vite)** `npm run client` • Portal UI: **http://localhost:5173** • Direct WebSocket connect in dev • Hot Module Replacement enabled

6. Pre-Seeded Test Login Credentials

Role	Email	Password	Account Scope & Dataset Highlights
Admin (Ops Desk)	`admin@bookleaf.com`	`admin123`	Full ticket queue triage, AI classification override, AI response draft synthesis, internal ops notes.
Author (Sample 1)	`priya.sharma@email.com`	`author123`	ID: AUTH001 • 2 live titles • ■11,970 earned • ■3,570 pending • ticket conversation.
Author (Sample 2)	`rohit.kapoor@email.com`	`author123`	ID: AUTH002 • 2 bestsellers • 2,000+ sold • global distribution channels.
Author (Sample 3)	`sneha.kulkarni@email.com`	`author123`	ID: AUTH007 • 2 live books + 1 in production (*Midnight in Mysore* - Cover Design).
All Authors	<firstname>.<lastname>@email.com	`author123`	All 10 authors (AUTH001 through AUTH010) are seeded with identical dataset attributes.

7. Technical Assumptions & Edge Cases Handled

• **Royalty Computation Rules:** Follows BookLeaf 80/20 royalty split (80% Author / 20% Publisher), quarterly payout cycles, and clearly flags whether pending royalties meet the ■1,000 disbursement threshold.
• **In-Production Titles:** Handled gracefully without zero-division or missing links (e.g., *Midnight in Mysore* and *Raising Roots* display dedicated production workflow milestone badges).
• **Real-Time Bi-Directional Chat:** Uses Socket.IO duplex communication. Author messages update admin queue immediately; staff replies reflect on author desks with zero page reloads.
• **Internal Note Security:** Private ops notes are strictly filtered out on author-facing endpoints using database-level Mongoose query projection (`.select('-internal_notes')`).
• **AI Fallback Resilience:** The AI classification and response generator incorporates an automated fallback to the internal BookLeaf Heuristic Engine so the app remains 100% functional without an external LLM key.